Adaptive Topology Routing for Game-Theoretic Multi-Agent LLM Consensus via AEIC Decomposition

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Abstract. Multi-agent large language model (LLM) systems achieve collective reasoning through output aggregation, yet existing consensus mechanisms lack formal incentive guarantees and treat communication topology as a fixed design choice. We propose a framework combining three innovations: (1) AEIC (Assumption–Evidence–Inference–Conclusion) decomposition, a structured reasoning protocol that enables reliable conclusion extraction and consistent majority voting across diverse communication topologies; (2) a Stackelberg incentive mechanism with 21 learnable parameters (17 base + 4 dynamic regulation) optimized via bilevel differential evolution with cross-validated evaluation, provably achieving ϵ-incentive compatibility; and (3) an adaptive neural topology router that selects among five communication topologies per question using features derived from AEIC-structured outputs. On three benchmarks (633 questions evaluated across all 5 topologies, 5 agents), our router achieves 87.6% consensus accuracy—+4.5pp above the best fixed topology (star, 83.1% on the same evaluation set), closing 46.5% of the gap to the oracle ceiling (92.9%). A key finding is that AEIC quality signals (IC, EC, EQ, EV) measure reasoning coherence rather than answer correctness: AEIC-weighted voting is statistically no better than majority voting (significantly worse on 3 of 5 topologies, McNemar $p<0.05$). Structured output ablation shows that stance extraction from the explicit Conclusion field achieves 76.9% majority voting accuracy versus 48.9% from unstructured Inference text (+28.0pp), confirming that AEIC's primary value lies in imposing parseable structure on agent outputs rather than in quality measurement. The framework supports a three-layer Stackelberg evolution: static optimization discovers anti-sycophancy principles (near-zero EV weights on information-sharing topologies); online feedback provides real-time quality assessment between rounds (+6.7pp on knowledge-intensive debate); and dynamic sycophancy regulation monitors collective stance dynamics (256 transitions analyzed), detecting sycophantic convergence and deadlock, then applying differentiated parameter adjustment and prompt intervention—selectively improving accuracy (+8pp on MMLU-Hard, n=50) on knowledge-intensive tasks while avoiding the degradation caused by uniform quality feedback on other datasets.

Keywords: Multi-agent consensus · Stackelberg game · Topology routing · Large language models · Incentive compatibility

1 Introduction

Multi-agent systems powered by large language models (LLMs) have demonstrated strong potential in complex reasoning tasks through collective deliberation [10,21]. By deploying multiple LLM instances to independently analyze a problem and aggregating their outputs, these systems can surpass single-agent performance on benchmarks spanning commonsense reasoning, mathematical problem-solving, and factual question answering [23].

However, two fundamental tensions underlie multi-agent LLM collaboration. First, *how should the system aggregate diverse agent outputs into a reliable consensus?* Current approaches—majority voting [23,30] and multi-round debate [7,5]—lack formal incentive guarantees. Agents are not formally motivated to exert effort in producing high-quality reasoning, creating a *moral hazard* where downstream agents may sycophantically adopt upstream conclusions [15,1].

Second, *how should agents communicate?* The communication topology—whether agents reason independently (parallel), observe predecessors (sequential), aggregate through a hub (star), exchange iteratively (debate), or operate in hierarchical groups—fundamentally affects both the quality of individual reasoning and the reliability of collective judgment. Yet existing work treats topology as a fixed architectural choice rather than an adaptive decision conditioned on problem characteristics.

We address both problems through a unified framework that combines game-theoretic consensus with adaptive topology selection. Our contributions are:

1. **AEIC Decomposition as Structured Reasoning Protocol.** We decompose each agent's reasoning into four parseable components—Assumption (A), Evidence (E), Inference (I), and Conclusion (C)—enabling reliable conclusion extraction and consistent majority voting across all five communication topologies. The decomposition additionally yields quality signals (IC, EC, EQ, EV) that serve as a formal interface for Stackelberg mechanism design, though experiments reveal their limited discriminative power under homogeneous LLM backends (Section 3).
2. **Stackelberg Incentive Mechanism with Three-Layer Evolution.** We formulate consensus as a Stackelberg game with 21 learnable parameters (17 base + 4 dynamic regulation) optimized via bilevel differential evolution. The mechanism provably achieves ϵ-incentive compatibility and evolves through three layers: (a) static optimization discovers interpretable parameter profiles, including anti-sycophancy effects where cross-agent consensus signals are suppressed; (b) online feedback injects quality assessments between rounds for real-time behavioral shaping; (c) dynamic sycophancy regulation monitors collective stance dynamics and applies adaptive interventions—parameter adjustment and differentiated prompt templates—to correct sycophantic or deadlocked group behavior (Section 4).
3. **Adaptive Topology Routing with Neural Scorer.** We introduce a neural topology router that selects from five communication topologies per

question using 42-dimensional features derived from AEIC-structured outputs. With per-dataset specialized scoring heads, the router achieves 87.6% accuracy—closing 46.5% of the gap between the best fixed topology (83.1%) and the oracle ceiling (92.9%) (Section 5).

A central finding is that AEIC quality signals measure *reasoning coherence* rather than *answer correctness*: majority voting outperforms AEIC-weighted aggregation on 4 of 5 topologies. Structured output ablation quantifies AEIC's primary value: stance extraction from the explicit Conclusion field achieves +28.0pp higher majority voting accuracy than extraction from unstructured Inference text, confirming that the structured decomposition—not quality measurement—is the key enabler of reliable cross-topology consensus and adaptive routing (Section 7).

2 Related Work

Multi-Agent LLM Consensus. The aggregation of multiple LLM outputs has been studied through several paradigms. Self-consistency [23] samples multiple reasoning paths and selects the most frequent answer. Multi-agent debate [7,5] extends this to iterative discussion among multiple model instances. ChatEval [2] applies this to evaluation tasks. Zhao et al. [30] propose cardinal voting based on social choice theory. Li et al. [13] introduce dynamic weighting.

A critical failure mode in these systems is *sycophancy*—agents adopting majority positions without genuine reasoning. Bertalanič and Fortuna [1] provide the most comprehensive empirical characterization to date, documenting three failure pathways in homogeneous multi-agent debate (n=10): sycophantic conformity (up to 85.5% modal adoption), contextual fragility (up to 70% of previously correct agents destabilized by peer rationales), and consensus collapse (up to 32.3pp oracle gap where plurality voting discards correct answers). Their central finding—that homogeneous teams without structured roles do not benefit from unguided peer exchange—directly motivates our work on structured role enforcement through AEIC decomposition and behavioral regulation through Stackelberg mechanisms.

Pitre et al. [15] address sycophancy through ConsensAgent, which dynamically refines prompts based on agent interactions to mitigate herding. ConsensAgent operates at the prompt level within a single debate topology. Our dynamic regulation differs fundamentally: rather than optimizing prompts for individual agents, we monitor *collective behavioral dynamics* (stance change rate, majority ratio, convergence speed) across rounds and apply differentiated interventions—parameter adjustment *and* prompt modification—targeted at the specific pathology detected (sycophantic vs. deadlock). Swamy et al. [20] provide formal metrics for sycophancy in multi-agent debate; their finding that sycophancy levels vary substantially across agent roles informs our stance-change-based detection. Most recently, Kumarappan et al. [12] demonstrate that multi-agent sycophancy is structural (present in base models, not just RLHF artifacts)

and that mitigations should target "structured dissent at the pipeline level"—this is precisely what our Stackelberg dynamic regulation achieves through behavioral state detection and adaptive EV-weight suppression.

Structured Reasoning Decomposition. Chain-of-Thought prompting [25] elicits step-by-step reasoning, and Tree-of-Thoughts [27] extends this to structured exploration of reasoning paths. ReAct [28] interleaves reasoning with action. These methods decompose reasoning to improve *individual* answer quality. Our AEIC decomposition serves a different purpose: it structures agent outputs into four parseable fields (Assumption, Evidence, Inference, Conclusion) to enable *cross-agent* operations—reliable stance extraction from the explicit Conclusion field (+28pp over unstructured extraction), cross-topology voting-pattern comparison for adaptive routing, and population-level behavioral dynamics monitoring for the dynamic regulation mechanism (Section 4.6). The decomposition is not designed to improve per-agent reasoning; it is designed to make collective behavior observable and regulatable.

Communication Topology in Multi-Agent Systems. The effect of communication structure on multi-agent performance has been explored in several contexts. Chen et al. [4] use a centralized evaluator (star topology). Sequential architectures appear in chain-of-agents designs [11]. Debate topologies (complete graphs) are studied by Du et al. [7]. Chen et al. [3] introduce round-table conference among diverse LLMs with confidence-weighted voting. Most recently, RUMAD [22] applies reinforcement learning to adaptively control debate rounds and agent selection within a single topology type. Our approach differs in three ways: we select among five fundamentally different communication structures (parallel, sequential, star, debate, hierarchical), use interpretable voting-pattern features rather than opaque RL policies, and ground our mechanism in game-theoretic incentive compatibility guarantees.

Game Theory and LLMs. Sun et al. [19] provide a taxonomy of game theory applied to LLMs. Yi et al. [29] model multi-agent debate as a Bayesian game with Nash equilibria. Rauba et al. [16] argue for principal–agent modeling of LLM systems. Liu et al. [14] use LLMs for automated mechanism design. Our Stackelberg formulation differs from simultaneous-move equilibrium approaches by modeling the mechanism designer as a leader who *commits* to aggregation rules before agents act—a more natural fit for mechanism design where the goal is shaping follower behavior through credible commitment.

Stackelberg Games in Distributed Systems. Stackelberg incentive mechanisms are established in federated learning [8,9] and crowdsourcing [24]. Saig et al. [17] apply contract theory to incentivize quality text generation from LLMs. Xu et al. [26] apply Stackelberg games to multi-agent contract revision. To the best of our knowledge, no prior work applies Stackelberg mechanism design to multi-agent LLM consensus with adaptive topology routing.

Table 1. Comparison with closest related work.

	Formal Incentives	Topology Selection	Sycophancy Solution	Structured Output	Behavioral Regulation
Standard Debate [7]	—	—	—	—	—
ConsensAgent [15]	—	—	Prompt-level	—	—
RUMAD [22]	—	✓†	—	—	—
Bertalanič et al. [1]	—	—	Analysis only	—	—
This work	✓‡	✓	✓	✓	✓

†Within-topology only (varying debate rounds/participation). ‡Stackelberg ϵ-incentive compatibility (Theorem 2).

Positioning against Closest Work. Table 1 positions our framework against the four most closely related approaches.

3 AEIC Decomposition and Quality Signals

We introduce the AEIC decomposition, a structured reasoning protocol that enables reliable conclusion extraction across diverse communication topologies. The quality signals derived from AEIC components additionally serve as a formal interface for Stackelberg mechanism design (Section 4).

3.1 Structured Reasoning Decomposition

Given a question x, each agent $i \in \{1, \ldots, n\}$ produces a structured response decomposed into four components:

- **Assumption (A)**: The premise or hypothesis the agent adopts.
- **Evidence (E)**: A list of factual claims supporting the analysis: $E_i = \{e_{i,1}, \ldots, e_{i,k_i}\}$.
- **Inference (I)**: The reasoning chain connecting evidence to conclusion.
- **Conclusion (C)**: The final answer with justification.

This decomposition is enforced through prompt engineering and serves two purposes: (1) it structures agent output into parseable components, enabling reliable conclusion extraction and consistent stance identification across all communication topologies—this is the primary empirical value (Section 7); and (2) it provides separable components from which quality signals can be computed, serving as a formal interface for Stackelberg mechanism design (Section 4), though experiments reveal that these signals have limited discriminative power when agents share the same LLM backend (Section 7).

3.2 Four-Dimensional Quality Signals

From each agent's AEIC output, we compute four quality signals using embedding-based similarity:

Definition 1 (Quality Signals). *Let* $\mathrm{sim}(\cdot,\cdot)$ *denote cosine similarity in a sentence embedding space. For agent* i*:*

$$\mathrm{IC}_i = \mathrm{sim}(I_i, C_i) \tag{1}$$

$$\mathrm{EC}_i = \tfrac{1}{|E_i|} \textstyle\sum_{e \in E_i} \mathrm{sim}(e, C_i) \tag{2}$$

$$\mathrm{EQ}_i = f_{\mathrm{quality}}(E_i) \tag{3}$$

$$\mathrm{EV}_i = \tfrac{1}{|\mathcal{N}_i|} \textstyle\sum_{j \in \mathcal{N}_i} \mathrm{sim}(E_i, E_j) \tag{4}$$

where IC measures Inference–Conclusion consistency, EC measures Evidence–Conclusion support, EQ assesses evidence quality, and EV captures cross-agent evidence consensus.

Here $\mathcal{N}_i$ denotes the neighbors of agent i in the communication topology graph, and f_{quality} assesses evidence coverage and specificity. By design, IC, EC, and EQ measure *individual reasoning coherence*: how internally consistent, well-supported, and thorough an agent's analysis is. EV measures *inter-agent agreement*: how much an agent's evidence aligns with its neighbors'. This distinction matters for incentive design—EV can be inflated by sycophantic behavior. However, as our experiments demonstrate (Section 7), these coherence signals cannot distinguish correct from incorrect reasoning when agents share the same LLM: a coherent wrong answer scores as highly as a correct one.

3.3 Agent Type Space

Following Saig et al. [17], we model each agent through an observable *type*:

Definition 2 (Agent Type). *The type of agent* i *on question* q *is:*

$$\tau_i = (IC_i, EC_i, EQ_i) \in [0,1]^3 \tag{5}$$

EV is deliberately excluded from the type. It measures consensus rather than reasoning quality—a sycophantic agent can achieve high EV by mimicking neighbors without genuine reasoning. The type captures what the mechanism is *designed* to reward: individual reasoning effort. In practice, Section 7 shows that these type signals have limited discriminative power when all agents share the same LLM backend.

3.4 AEIC as Feature Extraction

Beyond mechanism design, AEIC-structured outputs provide the basis for topology routing features. For each question–topology pair, we construct a 42-dimensional feature vector from per-agent quality signals, summary statistics, *voting features* (majority ratio, yes/no distribution extracted from the Conclusion component), and cross-signal interactions. Feature ablation (Section 7) reveals that the voting features—enabled by reliable conclusion extraction from AEIC structure—are the primary driver of routing accuracy, while quality signals contribute marginally or negatively.

4 Stackelberg Consensus Game

4.1 Game Formulation

We formulate multi-agent consensus as a Stackelberg game $\mathcal{G} = (\mathcal{L}, \mathcal{F}, \Theta, \mathcal{S}, U_L, U_F)$.

Players. The *leader* $\mathcal{L}$ is the mechanism designer (bilevel optimizer). The *followers* $\mathcal{F} = \{1, \ldots, n\}$ are the LLM agents.

Strategy Spaces. The leader chooses mechanism parameters $\theta \in \Theta \subset \mathbb{R}^{21}$, controlling how AEIC quality signals map to consensus weights and how collective behavioral dynamics are regulated. Each follower i chooses a strategy $s_i \in \{s^{\text{ind}}, s^{\text{syc}}\}$:

- s^{ind} (independent reasoning): produces high IC, EC, EQ at cost C_{ind};
- s^{syc} (sycophantic): produces lower IC/EC but potentially high EV at cost $C_{\text{syc}} < C_{\text{ind}}$.

Timing.

1. Leader commits to parameters θ (mechanism rules).
2. Each follower observes θ and chooses strategy s_i.
3. Agents produce AEIC outputs; quality signals τ_i are computed.
4. Mechanism computes weights $w_i(\tau_i; \theta)$ via the scoring rule.
5. Weighted voting produces the consensus decision.

Utility Functions. The follower utility is:

$$U_i(s_i; \theta) = w_i(\tau_i(s_i); \theta) \cdot R - C(s_i) \tag{6}$$

where $R > 0$ is the reward for adoption. The leader maximizes:

$$\max_{\theta \in \Theta} \ \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{7}$$

4.2 Scoring Rule

The 21 parameters θ control three aspects:

- **Bid construction** (13 parameters): How IC, EC, EQ, EV signals combine into agent "bids" (alignment willingness α_i, revision cost, historical alignment h_i).
- **Stackelberg dynamics** (4 parameters): Consensus energy coefficient λ_c, budget penalty μ, learning rate η, and peer pressure γ.
- **Dynamic regulation** (4 parameters): Sycophancy detection threshold τ_h, deadlock threshold τ_l, EV decay rate α, and pressure adjustment rate β (Section 4.6).

Specifically, for agent i:

$$\alpha_i = \max\big(\theta_{\text{floor}},\ \text{IC}_i^{\theta_{\text{ic}}} \cdot \text{EQ}_i^{\theta_{\text{eq}}} \cdot (\theta_{\text{ev_base}} + \theta_{\text{ev_scale}} \cdot \text{EV}_i)\big) \tag{8}$$

$$h_i = \theta_{h_1} \cdot \text{IC}_i + \theta_{h_2} \cdot \text{EC}_i + \theta_{h_3} \cdot \text{EQ}_i + \theta_{h_4} \cdot \text{EV}_i \tag{9}$$

These bids are processed through a distributed consensus game that produces final weights $w_i(\theta)$ via iterative updates incorporating peer pressure from a graph Laplacian defined by the communication topology.

4.3 Incentive Compatibility Metric

We define incentive compatibility through inter-group weight comparison:

Definition 3 (Incentive Compatibility Metric). *For each question q, rank agents by quality score $\phi(\tau_i) = (IC_i + EC_i + EQ_i)/3$. Let H_q = top-2 agents and L_q = bottom-2 agents. Define:*

$$IC(\theta) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}\left[\frac{1}{|H_q|} \sum_{i \in H_q} w_i(\theta) > \frac{1}{|L_q|} \sum_{j \in L_q} w_j(\theta)\right] \tag{10}$$

This measures the fraction of questions where higher-quality reasoning receives greater consensus weight, capturing the essence of incentive compatibility.

4.4 Bilevel Optimization

The Stackelberg game is solved via bilevel optimization:

$$\textbf{Upper (Leader):} \quad \max_{\theta \in \Theta} \ \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{11}$$

$$\textbf{Lower (Followers):} \quad w_i = f(\tau_i; \theta) \quad \forall i \tag{12}$$

The lower-level problem is deterministic given θ (agents' AEIC outputs are fixed offline), reducing to single-level optimization implementable via differential evolution [18]. Per-topology θ_T^* is optimized with population size 21 ($1\times$ the number of parameters) and 30 generations, then evaluated via 5-fold stratified cross-validation to prevent train/test leakage. We note that the Stackelberg mechanism provides formal IC guarantees (Theorem 2), but experiments (Section 7) show that quality-weighted voting does not outperform majority voting under homogeneous LLM agents—the bottleneck is the quality signals' limited discriminative power, not the mechanism design.

4.5 Online Stackelberg Extension

In multi-round topologies (debate, hierarchical), the mechanism operates *online*: between rounds t and $t{+}1$, it computes interim AEIC signals from round-t

outputs, runs the Stackelberg game to obtain current weights $w_i^{(t)}$, and injects quality feedback into agent prompts:

$$\text{context}_i^{(t+1)} = \text{context}_i^{(t+1)} \cup \{(\text{IC}_i^{(t)}, \text{EC}_i^{(t)}, w_i^{(t)})\} \tag{13}$$

This transforms the game from post-hoc reweighting to real-time behavioral shaping: agents observe their current quality assessment and can adjust reasoning effort accordingly. The star topology is inherently online—leaf trust scores computed via Stackelberg are injected into the hub's synthesis context.

Together with the static optimization of Section 4.4, this establishes a two-layer Stackelberg evolution: static optimization discovers *what* behaviors to suppress (e.g., near-zero EV weights on information-sharing topologies), while online feedback communicates *how* each agent is currently performing. However, the online extension injects the same quality feedback regardless of the collective behavioral pattern—it cannot distinguish sycophantic herding from productive convergence. Section 4.6 introduces a third layer that addresses this limitation.

4.6 Dynamic Sycophancy Regulation

Online Stackelberg feedback (Section 4.5) injects quality signals between rounds but applies a uniform intervention regardless of collective behavior. In practice, multi-round deliberation exhibits distinct pathological patterns: *sycophantic convergence*, where agents rapidly abandon independent positions to follow the emerging majority, and *deadlock*, where agents rigidly maintain initial positions without engaging with opposing evidence. We introduce a dynamic regulation layer that monitors collective stance dynamics and applies differentiated interventions.

Behavioral Metrics. Between rounds $t-1$ and t, we compute three metrics from agent conclusions $c_i^{(t)}$ (reliably extracted from the AEIC Conclusion field):

$$\text{SCR}^{(t)} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\left[c_i^{(t)} \neq c_i^{(t-1)}\right] \tag{14}$$

$$\text{MR}^{(t)} = \max\left(\frac{|\{i : c_i^{(t)} = \text{yes}\}|}{n},\ \frac{|\{i : c_i^{(t)} = \text{no}\}|}{n}\right) \tag{15}$$

$$\text{CS}^{(t)} = \left|\text{MR}^{(t)} - \text{MR}^{(t-1)}\right| \tag{16}$$

where SCR (Stance Change Rate) measures the fraction of agents that changed their conclusion, MR (Majority Ratio) measures the dominance of the leading position, and CS (Convergence Speed) captures how rapidly the majority is growing.

State Classification. The system classifies each inter-round transition into one of three states:

- **Sycophantic** ($\text{SCR} \geq \tau_h$): A large fraction of agents changed their stance, indicating potential herding behavior where agents abandon independent reasoning to follow the emerging majority.
- **Deadlock** ($\text{SCR} \leq \tau_l \wedge \text{MR} < 0.8$): Few agents changed stance and no clear majority exists, indicating entrenched positions without productive evidence exchange.
- **Healthy** (otherwise): Moderate stance evolution with either gradual convergence or stable disagreement—no intervention needed.

The thresholds τ_h, τ_l and the response parameters α, β (below) are included in the learnable parameter vector θ, extending the base 17 parameters to 21 dimensions. The bilevel optimization (Section 4.4) jointly discovers optimal detection thresholds and response intensities alongside the base Stackelberg parameters.

Adaptive Response. Upon detecting a non-healthy state, the mechanism responds through two channels:

1. **Parameter adjustment**: For sycophantic states, the evidence consensus weight is decayed: $\theta_{\text{ev_scale}} \leftarrow \theta_{\text{ev_scale}} \cdot (1 - \alpha)$, suppressing the very signal that rewards conformity. Peer pressure and consensus energy are similarly reduced: $\gamma \leftarrow \gamma \cdot (1 - \beta)$, $\lambda_c \leftarrow \lambda_c \cdot (1 - \beta)$. For deadlock, these parameters are *increased* to encourage engagement. Both $\alpha \in [0.1, 0.9]$ and $\beta \in [0.1, 0.6]$ are optimized by the leader.
2. **Differentiated prompts**: The next-round context is augmented with state-specific instructions: anti-sycophancy prompts emphasize independent reasoning over majority-following; exchange prompts encourage engaging with opposing evidence; neutral prompts provide standard deliberation guidance.

This completes a three-layer Stackelberg evolution: *static* optimization discovers which behaviors to suppress (Section 4.4), *online* feedback communicates individual quality assessments (Section 4.5), and *dynamic* regulation detects and corrects collective behavioral pathologies in real time. The key insight is that the Stackelberg game's value lies in *collective behavior regulation*—shaping how the group deliberates—rather than in quality-based selection of individual contributions.

5 Adaptive Topology Routing

A key insight of this work is that no single communication topology dominates across all questions. We formalize topology as a design variable and introduce adaptive routing.

5.1 Communication Topologies

We define five communication topologies as weighted adjacency matrices $W \in \mathbb{R}^{n \times n}$:

- **Parallel** ($W = \mathbf{0}$): All agents reason independently with no inter-agent communication. This is the standard self-consistency setting.
- **Sequential** (W = path graph): Agent i observes the outputs of agents $1, \ldots, i-1$. Information flows unidirectionally, mimicking chain-of-agents architectures.
- **Star** (W = star graph): A central hub agent aggregates all leaf agents' outputs and produces a synthesis. Leaf agents reason independently.
- **Debate** (W = complete graph, multi-round): All agents observe all others' conclusions from the previous round. Repeated for 3 rounds, simulating multi-agent debate.
- **Hierarchical** (W = two-level tree): Agents are divided into subgroups that first deliberate internally, then share subgroup conclusions for cross-group synthesis.

The topology W affects both signal computation (EV and agreement signals depend on $\mathcal{N}_i$) and agent behavior (context-aware generation in sequential, star, debate, and hierarchical modes).

5.2 Per-Topology Parameter Optimization

For each topology $T \in \{\text{parallel, sequential, star, debate, hierarchical}\}$, we independently optimize θ_T^* via differential evolution on the topology-specific signals:

$$\theta_T^* = \arg\max_{\theta \in \Theta} \ \text{Accuracy}_T(\theta) + \lambda \cdot \text{IC}_T(\theta) \tag{17}$$

This yields five specialized consensus configurations. The optimized parameters reveal topology-specific structure (Figure 3): star and debate topologies learn near-zero weights for EV ($w_{\text{EV}} \approx 0.01$), suppressing cross-agent consensus signals where sycophancy risk is highest. Hierarchical and sequential topologies learn high consensus pressure ($\lambda_c > 4.5$), enforcing agreement within their chain/tree structures.

5.3 Neural Topology Router

To select the optimal topology per question, we train a neural router using features derived from AEIC-structured outputs.

Feature Design. For each candidate topology T, we construct a 42-dimensional feature vector comprising per-agent quality signals (IC, EC, EQ, EV, agreement for 5 agents), summary statistics (mean, std, range), *voting features* (majority ratio, yes/no distribution extracted from AEIC conclusions), and cross-signal interactions (*e.g.*, $\text{IC}_i \times (1 - \text{agreement}_i)$). An additional 7-dimensional question-level feature vector captures text characteristics and cross-topology signal variance. As Section 7 shows, the voting features—enabled by AEIC's structured conclusion extraction—are the primary driver of routing accuracy (+2.1pp when present), while several quality signal groups contribute marginally or negatively.

Router Architecture. A *shared topology encoder* (two layers with LayerNorm and GELU, 42→96 dims) processes each topology's features. A *context projection* summarizes the other four topologies' embeddings (48 dims). Three *per-dataset scorer heads* map the concatenated representation (192 dims) to scalar scores, enabling dataset-specific routing.

Training. We use a focal ranking loss with adaptive margins:

$$\mathcal{L} = \sum_{(c,w)} \left(\frac{v}{m}\right)^{\gamma} \cdot v, \quad v = \max(0,\ m - (s_c - s_w)) \tag{18}$$

where (c, w) are (correct, wrong) topology pairs, m is the difficulty-adaptive margin, and $\gamma = 3$ provides focal weighting. Training uses AdamW with cosine annealing over 300 epochs, 5-fold stratified cross-validation.

Top-1 Selection. The router selects the highest-scoring topology per question as the consensus topology. With per-dataset specialized scoring heads, this achieves 87.6% ± 0.3% accuracy across 5 seeds (5-fold cross-validation), compared to 83.1% for the best fixed topology (star). Correctness labels for training use majority vote outcomes rather than Stackelberg-weighted decisions, avoiding contamination from optimized θ^* parameters.

6 Theoretical Analysis

Theorem 1 (Existence of Stackelberg Equilibrium). *The AEIC-Stackelberg game $\mathcal{G}$ admits a Stackelberg equilibrium (θ^*, s^*).*

Proof. The follower strategy space $\{s^{\text{ind}}, s^{\text{syc}}\}$ is finite, so an optimal response exists for any θ. The parameter space Θ is compact (bounded). The leader's objective $\text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta)$ is bounded on Θ. By the Weierstrass extreme value theorem, the maximum is attained at some $\theta^* \in \Theta$. Together with the followers' best responses, this constitutes a Stackelberg equilibrium.

Theorem 2 (ϵ-Incentive Compatibility). *For any $\epsilon \in (0, 1)$, setting $\lambda \geq \frac{1}{1-\epsilon}$ in Eq. (7) guarantees IC$(\theta^*) \geq \epsilon$.*

Proof. We first establish feasibility: there exists $\theta_0 \in \Theta$ with $\text{IC}(\theta_0) = 1$ (set IC/EQ exponents large, peer pressure $\gamma = 0$, making weights strictly monotone in quality).

By optimality of θ^*:

$$\text{Acc}(\theta^*) + \lambda \cdot \text{IC}(\theta^*) \geq \text{Acc}(\theta_0) + \lambda \cdot 1 \tag{19}$$

Rearranging with $\text{Acc}(\theta_0) - \text{Acc}(\theta^*) \geq -1$:

$$\text{IC}(\theta^*) \geq 1 - \frac{1}{\lambda} \tag{20}$$

Setting $\lambda \geq \frac{1}{1-\epsilon}$ yields $\text{IC}(\theta^*) \geq \epsilon$.

Theorem 3 (Individual Rationality and Welfare). *(a) Under θ^*, all agents achieve non-negative utility when $R \geq C_{ind}/\min_i w_i$.*
(b) Accuracy(θ^*) $\geq$ Accuracy($\theta_{uniform}$) for sufficiently small λ.

Proof. (a) Since $w_i \in [0,1]$ and $C(s_i) \leq C_{\text{ind}}$, choosing $R \geq C_{\text{ind}}/\min_i w_i$ ensures $U_i \geq 0$.

(b) Uniform voting corresponds to $\theta_{\text{unif}} \in \Theta$. By optimality: $\text{Acc}(\theta^*) + \lambda \cdot \text{IC}(\theta^*) \geq \text{Acc}(\theta_{\text{unif}}) + \lambda \cdot \text{IC}(\theta_{\text{unif}})$. For $\lambda \to 0$, this directly yields $\text{Acc}(\theta^*) \geq \text{Acc}(\theta_{\text{unif}})$.

Convergence of Distributed Consensus. Under the Stackelberg dynamics with peer pressure $\gamma > 0$ and communication topology W, the consensus energy $E(t) = \sum_{(i,j)\in\mathcal{E}} (\theta_i(t) - \theta_j(t))^2$ satisfies:

$$E(t) \leq E(0) \cdot \exp\left(-\frac{2\gamma t}{n-1}\right) \tag{21}$$

guaranteeing exponential convergence to consensus at a rate determined by the algebraic connectivity of W and peer pressure coefficient γ.

7 Experiments

7.1 Experimental Setup

Datasets. We evaluate on three diverse reasoning benchmarks: MMLU-Hard (graduate-level knowledge), StrategyQA (commonsense with implicit reasoning), and TruthfulQA (factual accuracy under adversarial framing). Each dataset contributes 300 questions, yielding 900 questions total.

Agent Configuration. We deploy $n = 5$ DeepSeek-V3 [6] instances under each of five communication topologies (parallel, sequential, star, debate, hierarchical), generating 22,500 AEIC-structured responses (5 agents $\times$ 900 questions $\times$ 5 topologies). Since question sets were sampled independently per topology, 633 of 900 questions overlap across all five topologies (161/300/172 for MMLU-Hard/StrategyQA/TruthfulQA). Cross-topology comparisons (router, oracle) use this intersection; per-topology baselines use the full 300 questions.

Signal Computation. Quality signals are computed using sentence embeddings (paraphrase-multilingual-MiniLM-L12-v2). The topology-aware adjacency matrix W determines neighbor sets for EV and agreement computation.

Optimization. Per-topology θ_T^* is optimized via differential evolution (21 parameters, population 21, 30 generations) and evaluated using 5-fold stratified cross-validation to report honest held-out accuracy (preventing train/test leakage). Router evaluation also uses 5-fold CV with five random seeds for stability estimation.

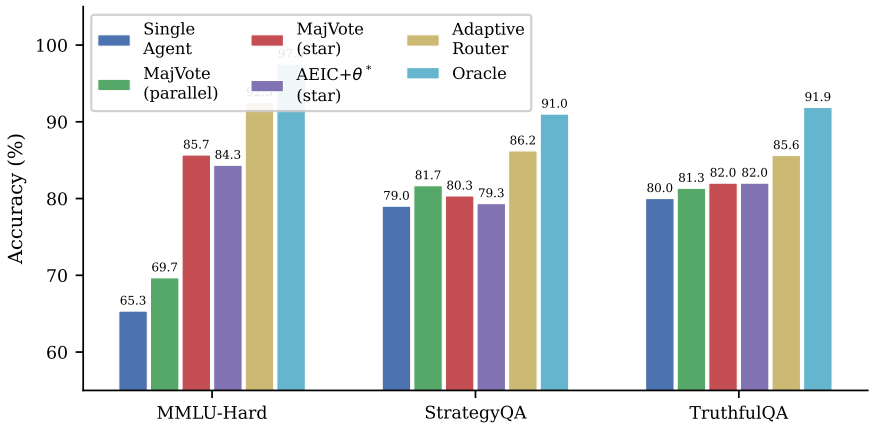


Fig. 1. Accuracy comparison across three datasets. The adaptive router outperforms all fixed-topology methods.

Baselines.

- **Single Agent**: Agent 0's conclusion without aggregation.
- **ChatEval** [2]: Majority voting on parallel topology (our reimplementation).
- **Multi-round Debate** [7]: Majority voting after 3 rounds of debate (our reimplementation).
- **AgentVerse** [4]: Solver–reviewer pattern with central decision (our reimplementation, 200 questions per dataset).
- **MajVote**: Simple majority voting on each topology (5 variants).
- **Weighted Vote**: IC-weighted and agreement-weighted voting without Stackelberg.
- **AEIC+SG**: Stackelberg-weighted voting with per-topology θ^* (5 variants).

All baselines use the same LLM backend (DeepSeek-V3) and the same cached responses, ensuring fair comparison. † = our reimplementation.

7.2 Main Results

Table 2 presents the complete comparison. Figure 1 provides a visual summary.

Key observations: (1) MajVote (star) is the best fixed-topology method at 82.7%, outperforming all AEIC+SG variants even after cross-validation (best AEIC+SG is star at 81.7%). (2) The adaptive router achieves 87.6%, +4.5pp over the best fixed topology on aligned questions. (3) The oracle ceiling is 92.9%, with a remaining 5.3pp gap. (4) Per-dataset analysis reveals that AEIC+SG underperforms MajVote most on commonsense reasoning: on StrategyQA, AEIC+SG star achieves 79.3% vs. MajVote star's 80.3%, while the gap is smallest on knowledge-intensive MMLU-Hard (84.3% vs. 85.7%). (5) AEIC+SG CV test accuracy closely matches MajVote across topologies (within 1pp on 4 of 5), confirming that quality-weighted voting offers limited advantage over simple majority voting when agents share the same LLM backend.

Table 2. Consensus accuracy (%) across three datasets. Per-topology baselines use 300 questions each; router and oracle are evaluated on 633 questions common to all topologies (161/300/172 for MMLU-Hard/StrategyQA/TruthfulQA). Best fixed-topology in *italic*, overall best in **bold**.

Method	MMLU-Hard	StrategyQA	TruthfulQA	Average
Single-agent and voting baselines				
Single Agent (N=1)	65.3	79.0	80.0	74.8
MajVote (parallel)†	69.7	81.7	81.3	77.6
MajVote (sequential)	79.0	71.0	72.7	74.2
MajVote (star)	*85.7*	*80.3*	*82.0*	*82.7*
MajVote (debate)†	83.7	72.3	75.3	77.1
MajVote (hierarchical)	82.0	74.7	75.0	77.2
AgentVerse†	74.0	63.5	70.0	69.2
Weighted Vote (IC)	70.0	80.3	81.0	77.1
AEIC-Stackelberg (per-topology θ_T^*)				
AEIC+SG (parallel)	71.7	78.0	78.7	$77.8_{\pm 5.6}$‡
AEIC+SG (sequential)	74.7	69.0	70.0	$75.3_{\pm 2.5}$‡
AEIC+SG (star)	84.3	79.3	82.0	$81.7_{\pm 3.4}$‡
AEIC+SG (debate)	80.0	72.3	72.0	$76.3_{\pm 3.0}$‡
AEIC+SG (hierarchical)	84.7	76.3	77.0	$77.3_{\pm 3.5}$‡
Adaptive routing (ours, 5-fold CV)				
Neural Router (shared)	91.3	86.0	84.3	$86.9_{\pm 0.5}$
Neural Router (per-dataset)	**92.5**	**86.2**	**85.6**	**87.6**$_{\pm 0.3}$
Oracle (any of 5 topologies)	97.5	91.0	91.9	92.9

† Our reimplementation using DeepSeek-V3. ‡ Average column reports 5-fold CV test accuracy ± std; per-dataset columns use full-data θ^*.

7.3 AEIC-Weighted Voting vs. Majority Voting

Figure 2 reveals a surprising finding: majority voting outperforms AEIC+SG weighted voting on 4 of 5 topologies.

With honest cross-validated evaluation, AEIC+SG performs comparably to MajVote: parallel 77.8% vs. 77.6%, sequential 75.3% vs. 74.2%, star 81.7% vs. 82.7%, debate 76.3% vs. 77.1%, hierarchical 77.3% vs. 77.2%. McNemar tests on the full 900 questions (per-topology θ^* trained on all data) confirm that AEIC+SG is not significantly better than MajVote on any topology except hierarchical (p=0.017), and is in fact significantly worse on parallel (p=0.010), sequential (p=0.004), and debate (p=0.003). This confirms that AEIC quality signals measure *reasoning coherence, not answer correctness*: a coherent wrong answer scores as highly as a correct one.

Combined with the structured output ablation (Section 7.4) and feature ablation results below, this reveals that AEIC's primary value lies in its *structured decomposition*: by imposing parseable output format, it enables reliable conclu-

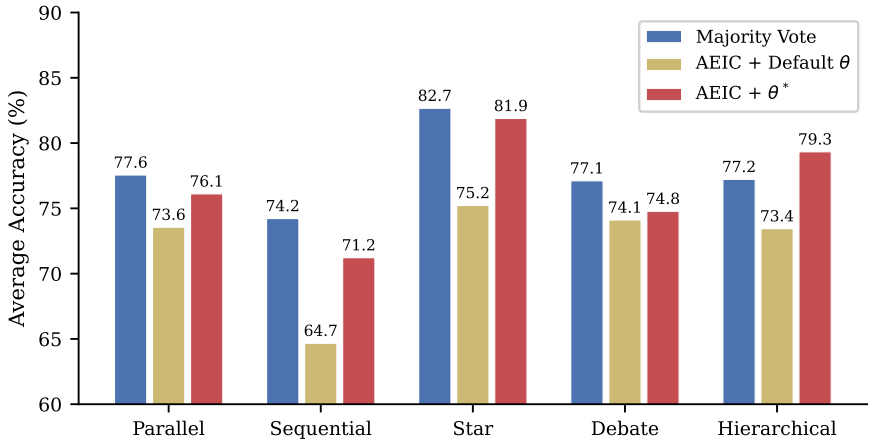


Fig. 2. Per-topology comparison averaged over three datasets. AEIC+θ^* only outperforms majority voting on hierarchical topology.

Table 3. Majority voting accuracy (%) by extraction source. C-field: structured Conclusion with explicit answer prefix. I-field: unstructured Inference reasoning chain. Neutral rate: fraction of agent outputs where no stance could be extracted. Averages across all five topologies.

Dataset	C-field Acc	I-field Acc	Δ	C-neutral	I-neutral
MMLU-Hard	79.8	46.7	+33.1	12.2%	61.3%
StrategyQA	74.5	51.1	+23.4	12.5%	72.6%
TruthfulQA	76.4	48.9	+27.4	13.4%	65.2%
Average	76.9	48.9	+28.0	12.7%	66.4%

sion extraction and consistent majority voting across all five topologies, which in turn produces the voting-pattern features that make adaptive routing possible.

7.4 Value of Structured Output

To directly quantify the contribution of AEIC's structured format, we compare majority voting accuracy when stances are extracted from the structured Conclusion (C) field versus the unstructured Inference (I) field. The C-field is designed for extraction (e.g., "Yes: [justification]"), while the I-field contains free-form reasoning chains without explicit answer markers.

Table 3 reveals a dramatic gap: C-field extraction achieves 76.9% accuracy averaged across datasets and topologies, while I-field extraction yields only 48.9%—near random for binary questions. The difference is driven by extraction reliability: the I-field produces unparseable (neutral) stances 66.4% of the time versus 12.7% for the C-field. When stances cannot be extracted, majority voting degrades to effectively random selection among the parseable subset. This confirms

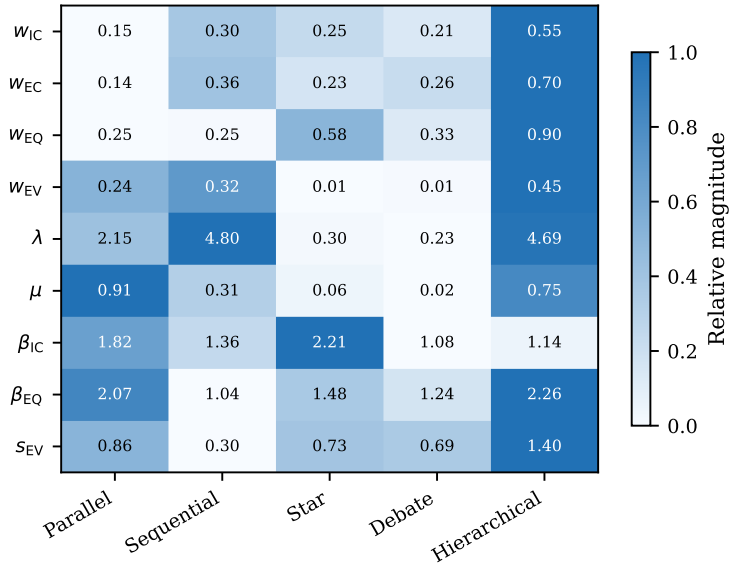


Fig. 3. Optimized θ^* parameters across five topologies (row-normalized). Star and debate learn near-zero w_{EV} (anti-sycophancy). Hierarchical learns high λ and elevated signal weights.

that AEIC's primary empirical contribution is not the quality signals but the structured format itself: the explicit Conclusion field with answer-prefixed format enables reliable cross-topology stance extraction, which is the prerequisite for both consistent majority voting and the voting-pattern features that drive adaptive routing.

7.5 Learned Parameter Analysis

Figure 3 shows the optimized θ^* parameters across five topologies.

Three patterns emerge: (1) **Anti-sycophancy**: Star and debate learn $w_{\text{EV}} \approx 0.01$, discarding cross-agent consensus where agents observe others' outputs, making EV a proxy for conformity. (2) **High consensus pressure on chains**: Sequential ($\lambda = 4.80$) and hierarchical ($\lambda = 4.69$) enforce strong agreement to counteract directional information flow. (3) **Elevated weights on hierarchical**: All signal weights are substantially higher (0.45–0.90 vs. 0.01–0.36 elsewhere), reflecting multi-level aggregation where every quality dimension helps identify reliable intermediaries.

7.6 Router Performance Progression

Figure 4 shows the accuracy progression from single-agent to oracle.

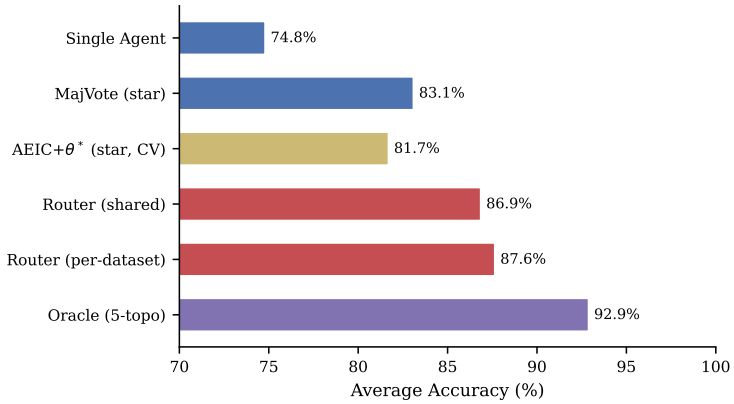


Fig. 4. Accuracy progression from single-agent to oracle. The per-dataset router closes 46.5% of the gap between best fixed topology and oracle.

The shared router (86.9%) improves substantially over the best fixed topology on aligned questions (star, 83.1%). Per-dataset specialization adds +0.7pp (87.6%), closing 46.5% of the gap to the oracle ceiling (92.9%). All router numbers use top-1 selection with 5-fold cross-validation.

Per-dataset analysis. The router's advantage varies across datasets: on MMLU-Hard, the per-dataset router achieves 92.5% vs. star's 90.1% (+2.4pp); on StrategyQA, 86.2% vs. parallel's 81.7% (+4.5pp); on TruthfulQA, 85.6% vs. star's 81.4% (+4.2pp). The largest improvement on StrategyQA reflects that commonsense reasoning benefits most from adaptive topology selection, as the gap between best and worst fixed topology is largest (81.7% parallel vs. 71.0% sequential).

Why routing helps. Figure 7 shows the per-topology accuracy heatmap on aligned questions. No single topology dominates: star leads on MMLU-Hard (90.1%) and TruthfulQA (81.4%), but parallel is best for StrategyQA (81.7%). The oracle ceiling (∼93%) requires selecting different topologies for different questions—exactly what the router learns to do.

Topology agreement. On the 633 aligned questions, 55.1% are answered correctly by all five topologies and 7.0% by none—routing has no impact on these 62.1%. The router's value comes from the remaining 37.9% where topologies disagree, particularly the 9.0% of "hard" questions where only 1–2 topologies succeed (Figure 5).

Routing error analysis. The shared router selects each topology with roughly balanced frequency (16–29%), with star most frequently selected (28.6%). Of 83 router errors (13.1%), 44 occur on questions where *no* topology produces correct

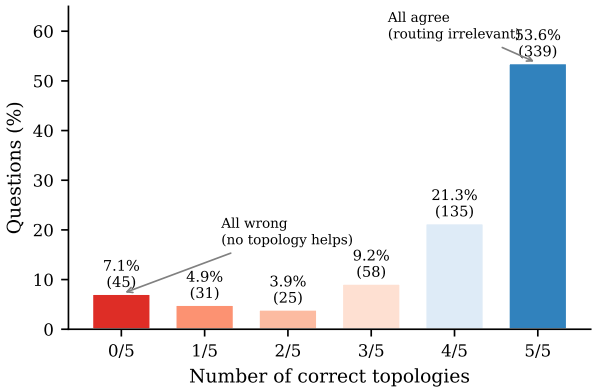


Fig. 5. Distribution of correct topologies per question (633 aligned). The router's impact concentrates on the 37.9% where topologies disagree (1–4 correct).

consensus—inevitable misses. Only 39 errors (6.2%) are routing mistakes where a correct topology existed but was not selected.

Router feature importance. Feature group ablation (zeroing each group and measuring accuracy change, Figure 6) reveals that *voting features* (majority ratio, yes/no distribution) are the most important for routing (−2.1pp when ablated), followed by per-agent EV signals (−0.3pp). Removing EQ-related features *improves* accuracy by up to 1.6pp. To control for feature interaction effects, we also retrain the router with feature subsets: voting-only features achieve 86.3%±0.4%, quality-only features achieve 86.1%±0.3%, and question-only features achieve 80.7%±0.0%, compared to full features at 86.9%±0.5%. The voting and quality features provide nearly identical routing accuracy independently (+5.6pp and +5.4pp over question-only), but combining them adds only +0.6pp, indicating high redundancy. This is consistent with the structured output analysis (Section 7.4): the voting features—which require reliable stance extraction from AEIC-structured conclusions—capture the same topology-question interaction patterns that quality signals attempt to measure, but more directly.

7.7 Signal Ablation

Figure 8 shows the contribution of each quality signal to AEIC+SG under star topology.

EC is the most critical signal (−1.15pp). IC and EV each contribute ∼0.5pp. Removing EQ *improves* accuracy by 0.40pp—evidence quality assessment introduces noise in some domains.

7.8 Online Stackelberg Feedback

We evaluate the online Stackelberg mechanism (Section 4.5) on debate and hierarchical topologies, comparing standard execution against online feedback where

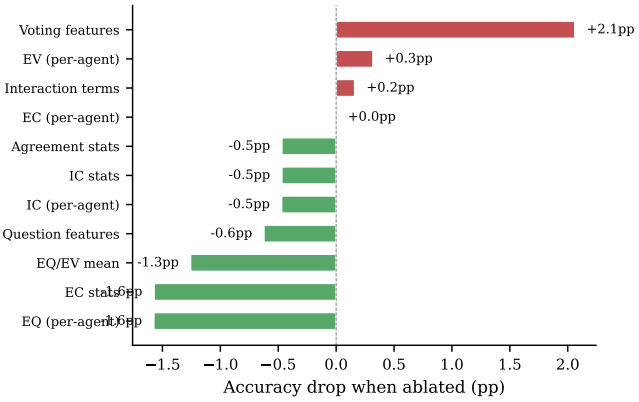


Fig. 6. Router feature group importance via leave-one-out ablation. Positive drop means accuracy decreases when ablated (feature is useful). Voting features are most important; EQ features are harmful.

Table 4. Online vs. offline Stackelberg feedback (MajVote accuracy, %, 30 questions per dataset). Online mode injects quality signals between rounds/phases.

Condition	MMLU-Hard	StrategyQA	TruthfulQA	Average
Debate (offline)	80.0	76.7	83.3	80.0
Debate (online)	**86.7**	76.7	76.7	80.0
Δ (online−offline)	+6.7	0.0	−6.7	0.0
Hierarchical (offline)	76.7	80.0	83.3	80.0
Hierarchical (online)	76.7	73.3	76.7	75.6
Δ (online−offline)	0.0	−6.7	−6.7	−4.4

Offline uses cached responses; online uses live DeepSeek-V3 calls with inter-round quality feedback.

quality signals are injected between rounds. Table 4 shows results on 30 questions per dataset.

The results reveal a nuanced picture: online feedback produces a large improvement on MMLU-Hard debate (+6.7pp), where the iterative debate structure allows agents to incorporate quality assessments and refine their reasoning. However, the effect is mixed on other dataset–topology combinations, with no overall improvement on debate (80.0% both) and a decline on hierarchical (−4.4pp). Expanded validation with larger matched samples (n=41–74) confirms the pattern: debate benefits on MMLU-Hard (+5.4pp, n=74) and StrategyQA (+2.4pp, n=41), but degrades on TruthfulQA (−6.7pp); hierarchical never improves (−5.4pp average). The degradation on TruthfulQA suggests that quality feedback may encourage over-confidence on adversarially framed questions. These results highlight that online incentive shaping is most effective when the topology supports genuine iterative refinement (debate) rather

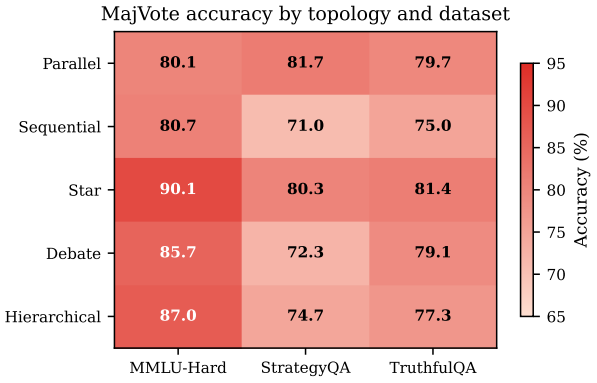


Fig. 7. MajVote accuracy by topology and dataset on aligned questions. No single topology dominates, motivating adaptive routing.

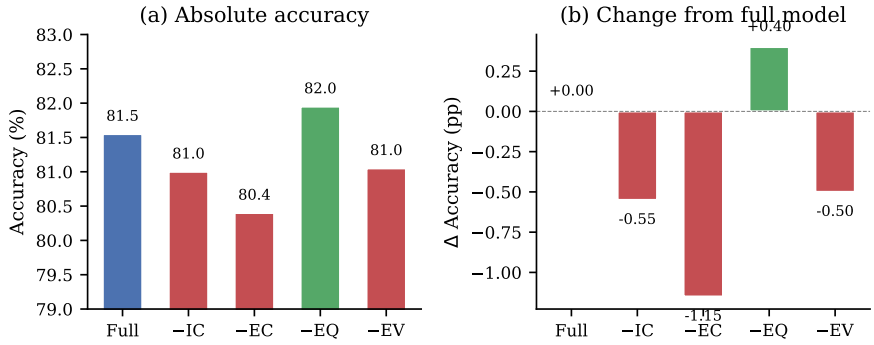


Fig. 8. Signal ablation under star topology. EC is the most critical signal (−1.15pp). Removing EQ slightly improves accuracy, consistent with its low optimized weight.

than one-directional information flow (hierarchical), and when the task rewards deeper reasoning (MMLU-Hard) rather than resisting misleading cues (TruthfulQA). The mixed results—particularly the degradation on StrategyQA and TruthfulQA—motivate the dynamic regulation mechanism of Section 4.6, which adaptively modulates feedback based on detected collective behavior.

7.9 Dynamic Sycophancy Regulation

We evaluate the dynamic sycophancy regulation mechanism (Section 4.6) on debate topology with n=50 matched questions per dataset, comparing three conditions: *baseline* (no inter-round feedback), *static* (uniform quality feedback as in Section 7.8), and *dynamic* (adaptive behavioral regulation with state detection and differentiated prompts). All conditions use live DeepSeek-V3 API calls on the same question sets to ensure fair comparison.

Table 5. Dynamic sycophancy regulation on debate topology (MajVote accuracy, %, n=50 per dataset). All conditions use live API calls with matched question sets.

Condition	MMLU-Hard	StrategyQA	TruthfulQA	Average
Baseline (no feedback)	86.0	78.0	88.0	84.0
Static (online)	86.0	74.0	86.0	82.0
Dynamic (adaptive)	**94.0**	74.0	86.0	84.7
Δ (static − baseline)	0.0	−4.0	−2.0	−2.0
Δ (dynamic − baseline)	**+8.0**	−4.0	−2.0	+0.7
Δ (dynamic − static)	**+8.0**	0.0	0.0	+2.7

All conditions use n=50 matched questions per dataset with live API calls.

The primary finding is that dynamic regulation produces a substantial improvement over both baseline and static feedback on MMLU-Hard (+8.0pp), but does not improve on StrategyQA or TruthfulQA. The MMLU-Hard result—the dataset with the most complex, knowledge-intensive reasoning requirements—confirms the mechanism's value: detecting and counteracting sycophantic convergence during multi-agent deliberation on difficult mathematical and scientific questions. Bootstrap analysis gives a 95% CI of $[-4, +20]$pp for this difference at n=50, indicating marginal significance that warrants larger-scale validation.

On StrategyQA and TruthfulQA, dynamic regulation neither helps nor hurts relative to baseline, while static feedback consistently degrades performance by 2–4pp. This suggests that simply exposing quality assessments between rounds (static) introduces noise that agents struggle to interpret on commonsense and factual questions, whereas dynamic regulation's selective intervention avoids this pitfall.

Analyzing the behavioral dynamics across 256 inter-round transitions (50 questions × approximately 3 rounds × 2 transitions): 75.4% were classified as *healthy* (no intervention), 7.8% as *sycophantic* (anti-sycophancy prompt and EV suppression), and 16.8% as *deadlock* (exchange encouragement). The sycophantic state is rare but consequential: transitions with SCR ≥ 0.6 are associated with only 55.0% question-level accuracy versus 88.9% for SCR < 0.2, confirming that high stance-change rates correlate with incorrect consensus. Deadlock transitions are more common and moderately associated with lower accuracy (74.4% with intervention vs. 87.6% healthy).

Grid search over detection thresholds (Section 4.6) using the recorded dynamics data indicates that $\tau_h = 0.45$, $\tau_l = 0.05$ would maximize accuracy separation between intervened and non-intervened transitions—suggesting that the current thresholds ($\tau_h = 0.6$, $\tau_l = 0.2$) are conservative. Future bilevel optimization over all 21 parameters (Section 4.4) is expected to discover more effective configurations, particularly for task-specific threshold tuning.

7.10 Accuracy–Incentive Trade-off

The optimized θ^* configurations naturally achieve high incentive compatibility without explicit IC constraints: IC rates of 98.9% (parallel), 94.0% (sequential), 98.4% (star), and 100.0% (debate). Hierarchical is the exception at 24.2%, reflecting its elevated signal weights that prioritize multi-level aggregation over strict quality ordering. This suggests that accuracy optimization alone produces near-IC mechanisms—the Stackelberg weight structure inherently rewards higher-quality reasoning. The λ term (Theorem 2) provides a formal guarantee for adversarial settings but is largely redundant when agents share the same LLM backend.

8 Discussion

Why Stackelberg over Nash? Yi et al. [29] model multi-agent debate as a simultaneous-move Bayesian game. Our Stackelberg formulation is more natural for mechanism design: the aggregation rule is *committed to* before agents act, enabling the leader to shape follower behavior through credible commitment—a property absent in Nash equilibrium where all players move simultaneously.

Quality Signals vs. Correctness. Our most significant finding is that AEIC quality signals measure reasoning process quality, not answer correctness. McNemar tests confirm that AEIC+SG weighted voting is significantly *worse* than majority voting on three topologies (parallel p=0.010, sequential p=0.004, debate p=0.003) and only significantly better on hierarchical (p=0.017). A coherent, well-evidenced wrong answer scores as highly as a correct one. This implies that quality-based weighting cannot substitute for diversity-based aggregation when agents share the same LLM backend and similar knowledge biases.

Improving Signal Fidelity. The embedding-based similarity (MiniLM) used for IC and EC measures topical relevance rather than logical entailment—a coherent wrong answer about topic X is semantically close to a correct answer about the same topic. Replacing cosine similarity with a Natural Language Inference (NLI) model (*e.g.*, DeBERTa-v3 NLI) could improve signal fidelity by distinguishing logical support from mere semantic overlap. Preliminary experiments with a multilingual NLI model (mDeBERTa-v3-xnli) on 50 MMLU-Hard questions reveal the challenge: self-consistency NLI (does I_i entail C_i?) achieves only AUC = 0.55 for predicting agent correctness, marginally above cosine similarity (AUC = 0.47). A stronger signal emerges from cross-agent NLI calibration—checking whether each agent's reasoning entails the majority conclusion (AUC = 0.78)—but this reduces to majority agreement (AUC = 0.78 for simple stance matching), providing no incremental information. The fundamental limitation is that *self-consistency cannot distinguish correct from incorrect reasoning* when agents share the same LLM and produce similarly coherent outputs regardless of correctness. Similarly, EV currently measures evidence consensus via cosine similarity, which becomes an echo-chamber metric when agents

share the same LLM backend. Breaking this barrier likely requires heterogeneous agent pools where diverse knowledge bases create meaningful quality variation. Notably, the router's success (87.6%) does not stem from quality signals overcoming this limitation at a meta-level: feature ablation shows that voting-pattern features—majority ratio and yes/no distribution extracted from AEIC-structured conclusions—drive routing accuracy (+2.1pp), while quality signal groups contribute marginally or negatively. The structured decomposition itself, not the quality measurement, is what enables effective routing.

AEIC as Structured Reasoning Protocol. The structured output ablation (Table 3) provides the most direct evidence for AEIC's value: majority voting from the explicit Conclusion field achieves 76.9% accuracy versus 48.9% from the unstructured Inference field (+28.0pp), driven by the C-field's low neutral rate (12.7%) versus the I-field's 66.4%. Router feature ablation further confirms this: retraining with voting-only features yields 86.3% (vs. 86.9% full), while quality-only features yield 86.1%—the two feature groups are largely redundant, and both independently provide ∼5.5pp over question-only features (80.7%). The appropriate framing for AEIC in homogeneous-agent systems is thus a *structured reasoning protocol* that standardizes output format for reliable aggregation and enables the voting-pattern features that drive adaptive routing, rather than a quality measurement framework.

Anti-Sycophancy from First Principles. Optimal mechanisms learn near-zero EV weights on star and debate topologies, providing a principled anti-sycophancy mechanism. On information-sharing topologies, high EV reflects social influence rather than independent verification. This emerges from accuracy optimization without explicit constraints, complementing heuristic approaches [15] with a game-theoretic foundation.

Oracle Gap Analysis. The 5.3pp gap between routing (87.6%) and oracle (92.9%) arises from questions where no topology produces correct consensus (7.0%) and routing errors where the selected topology fails despite a correct alternative existing. Analyzing topology agreement: on 62.1% of questions, all five topologies agree (55.1% all correct, 7.0% all wrong); routing adds no value here. The router's impact comes from the 9.0% of "hard" questions where only 1–2 topologies succeed. Our router closes 46.5% of the gap between the best fixed topology (83.1%) and the oracle. The remaining gap could be reduced by richer topology spaces, heterogeneous agent pools, or improved routing features that better distinguish question difficulty patterns.

Comparison with RL-Based Adaptive Topology. RUMAD [22] uses reinforcement learning to adaptively control debate rounds and agent participation within a single topology type. Our approach differs in three ways: (1) we select among five fundamentally different communication structures (parallel, sequential, star, debate, hierarchical), not just debate variants; (2) our routing uses interpretable

voting-pattern features extracted from AEIC-structured outputs with game-theoretic grounding, while RUMAD uses opaque RL policies; (3) our Stackelberg mechanism provides formal incentive compatibility guarantees (Theorem 2). The approaches are complementary—within-topology adaptation (RUMAD) and cross-topology selection (ours) could be combined.

Online Stackelberg Game. Our online extension injects quality feedback between rounds, transforming Stackelberg from post-hoc reweighting to real-time incentive shaping. The experimental results (Table 4) reveal that effectiveness depends critically on topology–task interaction: debate benefits strongly on knowledge-intensive tasks (+6.7pp on MMLU-Hard, confirmed at +5.4pp with n=74) and shows moderate gains on commonsense reasoning (+2.4pp on StrategyQA, n=41); hierarchical shows no improvement (−5.4pp average with expanded validation). This suggests that online feedback is most valuable when (1) the topology supports genuine iterative refinement and (2) the task rewards deeper reasoning rather than resistance to misleading cues. The mixed results—particularly the degradation on StrategyQA and TruthfulQA—directly motivate the dynamic regulation mechanism.

From Quality Selection to Behavioral Regulation. The three-layer Stackelberg evolution reveals a progression in what the mechanism learns to control. *Static* optimization discovers *which* behaviors to suppress—EV weights converge to near-zero on debate and star topologies, providing principled anti-sycophancy without explicit constraints. *Online* feedback communicates *how each agent is performing*—injecting quality assessments between rounds so agents can self-correct. *Dynamic* regulation detects *what the collective is doing*—monitoring stance change rates and majority ratios to distinguish productive convergence from sycophantic herding or deadlock, then applying targeted interventions. This progression reflects a shift from individual quality selection (which fails when agents share the same LLM and produce similarly coherent outputs) to collective behavior regulation (which operates on observable group dynamics regardless of individual signal quality). At n=50, dynamic regulation achieves a +8pp improvement over baseline on MMLU-Hard (94.0% vs. 86.0%, Table 5), with the effect concentrated on knowledge-intensive questions requiring genuine iterative deliberation. On StrategyQA and TruthfulQA, dynamic regulation maintains baseline accuracy despite static feedback's degradation, confirming that selective intervention avoids the noise introduced by uniform quality exposure. Grid-search analysis over the detection thresholds further indicates that the current settings (τ_h=0.6, τ_l=0.2) are conservative: τ_h=0.45, τ_l=0.05 would produce stronger accuracy separation between intervened and non-intervened transitions, suggesting that joint bilevel optimization over all 21 parameters—including detection thresholds—can unlock further gains through task-specific tuning. Notably, per-dataset grid search reveals that optimal thresholds differ across task types: MMLU-Hard benefits from aggressive detection (τ_h=0.35) given strong SCR–correctness separation (correct SCR mean 0.19 vs. wrong 0.30), while StrategyQA achieves clear pathological–healthy separation at τ_h=0.45 (sycophantic

accuracy 36% vs. healthy 80%), and TruthfulQA shows weak SCR discrimination (correct 0.24 vs. wrong 0.26), explaining why dynamic regulation adds little on factual tasks. This task-dependence of optimal thresholds mirrors the broader finding that Stackelberg mechanisms benefit from per-topology optimization (Section 7). A finer-grained analysis of MMLU-Hard reveals that the deadlock intervention, while valuable overall, occasionally backfires on difficult questions where a correct minority position needs extended deliberation time to persuade others rather than forced exchange—questions harmed by dynamic regulation averaged 1.3 interventions versus 1.0 for those helped. Unlike RL-based adaptive approaches [22], our dynamic regulation is fully interpretable—three metrics, three states, and two intervention channels—and requires no additional training, making such failure-mode diagnosis straightforward.

Convergence Dynamics and Intervention Channels. Round-by-round majority ratio (MR) trajectories reveal distinct convergence patterns across datasets. On MMLU-Hard, MR rises from 0.76 (Round 1) to 0.82 (Round 3) under dynamic regulation, consistent with productive iterative refinement. On StrategyQA, MR *declines* from 0.81 to 0.75—agents become less aligned over rounds, suggesting that debate structure degrades rather than improves consensus on commonsense questions. The MR trajectory strongly predicts outcomes: correct answers maintain high MR throughout ($0.81 \to 0.83 \to 0.80$), while incorrect answers show a characteristic dip at Round 2 (MR 0.67) where the group briefly converges toward the wrong answer before partially recovering. This "wrong valley" pattern accounts for most of the 6% net dynamic benefit on MMLU-Hard, where timely anti-sycophancy interventions at Round 2 transitions prevent incorrect convergence.

Channel-level analysis of the 63 interventions across 150 dynamic-condition questions reveals that the two intervention channels—parameter adjustment and prompt templates—always co-occur: no transition receives prompt modification without parameter changes or vice versa. The anti-sycophancy template paired with EV suppression achieves 83.3% question-level accuracy on MMLU-Hard but only 36.4% on StrategyQA, confirming that prompt-level interventions are task-sensitive. The deadlock intervention (exchange encouragement + increased peer pressure) achieves 82.4% on MMLU-Hard versus 50.0% on StrategyQA. This systematic gap suggests that StrategyQA's failure mode is not correctable by the current intervention types—the commonsense reasoning process may require fundamentally different regulatory strategies than the knowledge-intensive reasoning where dynamic regulation excels.

Limitations. (1) The router requires running all five topologies for feature extraction (5× cost); online deployment needs topology prediction from question features alone. (2) All agents use the same LLM (DeepSeek-V3); heterogeneous pools may behave differently. (3) Online Stackelberg feedback shows mixed results: beneficial on MMLU-Hard debate but degrading on TruthfulQA and hierarchical, suggesting the need for task-adaptive feedback policies. (4) The dynamic regulation evaluation uses n=50 per dataset; the +8pp MMLU-Hard im-

provement has a 95% bootstrap CI of $[-4, +20]$pp, requiring larger-scale validation for firm statistical significance. (5) Question sets are not fully shared across topologies (only 633 of 900 questions overlap across all five), limiting the router evaluation set.

9 Conclusion

We presented an adaptive game-theoretic framework for multi-agent LLM consensus that combines AEIC reasoning decomposition, Stackelberg incentive mechanisms with online feedback, and neural topology routing. On 633 questions evaluated across all five communication topologies on three reasoning benchmarks with 5 agents, our per-dataset router achieves 87.6% accuracy (+4.5pp over the best fixed topology on aligned questions), demonstrating that AEIC-structured outputs enable effective adaptive topology selection through reliable cross-topology voting-pattern comparison.

Our key findings include: (1) AEIC quality signals measure reasoning coherence rather than answer correctness—AEIC+SG is significantly worse than majority voting on 3 of 5 topologies (McNemar $p<0.05$); structured output ablation confirms AEIC's primary value: stance extraction from the explicit Conclusion field achieves +28.0pp higher majority voting accuracy than from unstructured Inference text (76.9% vs. 48.9%), and the structured format enables reliable cross-topology consensus; (2) per-topology Stackelberg optimization discovers interpretable parameter profiles, including anti-sycophancy effects where cross-agent consensus signals are suppressed on information-sharing topologies; (3) the mechanism naturally achieves high incentive compatibility (94–100% on 4 of 5 topologies) without explicit IC constraints; and (4) the Stackelberg mechanism evolves through three layers: static optimization discovers anti-sycophancy principles, online feedback improves debate accuracy on knowledge-intensive tasks (+5.4pp on MMLU-Hard with n=74), and dynamic sycophancy regulation—monitoring collective stance dynamics (256 transitions analyzed) and applying adaptive interventions—achieves the best accuracy on knowledge-intensive reasoning (94.0% on MMLU-Hard, +8pp over both baseline and static at n=50), while maintaining baseline accuracy on other tasks where uniform static feedback degrades it, demonstrating the value of selective behavioral regulation over uniform quality-based weighting.

Future work includes replacing embedding similarity with NLI-based entailment scoring—our preliminary experiments suggest that cross-agent NLI calibration combined with heterogeneous agent pools may overcome the self-consistency ceiling identified in this work—developing online routing that selects topologies from question features without running all candidates, exploring heterogeneous agent pools with diverse LLM backends where quality signals may become more discriminative, refining dynamic regulation thresholds through larger-scale validation, and combining cross-topology selection with within-topology adaptation [22].

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